TED(10)5008
(REVISION 2010)

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING TECHNOLOGY
Model question paper
CONCRETE TECHNOLOGY

Maximum marks (100) Time: 3 hours.

PART A

I (Answer the following questions in one or two sentences. Each question carries 2 marks.) Marks.

1 What is meant by hydration of cement?
2 What is meant by creep?
3 Define mix design.
4what is no-fines concrete?
5 What is carbonation? (5x2=10)

PART-B

II (Answer any five of the following questions. Each question carries 6 marks)

a. Explain the procedure to find out standard consistency of cement.
b. Explain the effect of retarding admixtures on concrete.
c. Explain any five factors affecting workability of concrete.
d. Explain Vee-Bee consistometer test.
e. What are the variable factors to be considered in connection with specifying a concrete mix.
f. Explain self compacting concrete.
g. What are the remedial measures to be taken against corrosion of cement. (5x6=30)

PART-C

(Answer one full question from each unit. Each question carries 15 marks)

UNIT-1

III a. Explain soundness test. 8

b. Explain surface texture and its influence on flexural strength of concrete. 7

OR

1V a. Explain flakiness index and elongation index. 10

b. Explain alkali aggregate reaction. 5

UNIT-11

V a. What are the Properties of fresh Concrete. 8

b. Explain membrane curing. 7

OR

V1 a. Explain the terms.1.seggregation, 2.bleeding.3) batching. 9

b. Explain flexural strength of concrete. 6

UNIT-111

V11. Explain the mix design procedure based on IS 10262-1982. 15

OR

V111 a. Explain any four terminologies that are used in the statistical quality Control of concrete. 10

b. Explain target strength for mix design. 5

UNIT-1V

1X. Explain the preparation of sulphur-infiltrated concrete and its applications. 15

OR

X. Explain the special problems encountered in hot weather concreting. 15